BEING LOCALLY THE IMAGE OF A DUALIZING COMPLEX IS INDEPENDENT OF THE AFFINE OPEN COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A86 (sources/stacks-duality/pinned-original-source.tex line 308).

Allowed standard prerequisites (editorial).

- Comparison of complexes on an affine scheme with complexes of modules. Derived Categories of Schemes, Lemma labelled perfect-lemma-affine-compare-bounded, cited at proof lines 324-328 and 340-344. Produces omega_A and identifies the localisations.
- Standard open coverings subordinate to a given affine cover. Schemes, Lemma labelled schemes-lemma-standard-open-two-affines, cited at proof lines 333-337. Provides the D(f_j) inside the U_i.
- Localisation and glueing of dualizing complexes. Dualizing Complexes, Lemmas labelled dualizing-lemma-dualizing-localize and dualizing-lemma-dualizing-glue, cited at proof lines 348-352. Give dualizingness on the localisations and then for omega_A.

Source reference. Original lemma 06Z0: sources/stacks-chow/tags.txt, source lines 8904–8904. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 329 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01IW: sources/stacks-chow/tags.txt, source lines 1993–1993. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 337 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A7G: sources/stacks-chow/tags.txt, source lines 12879–12879. Cross-chapter standard prerequisite located in the project tag list

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staged in the local snapshot; invoked at proof line 346 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A7H: sources/stacks-chow/tags.txt, source lines 12880–12880. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 351 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let K be an object of $D(\mathcal{O}_X)$. The following are equivalent*

(1) *For every affine open $U = \mathrm{Spec}(A) \subset X$ there exists a dualizing complex $\omega_A^\bullet$ for A such that $K|_U$ is isomorphic to the image of $\omega_A^\bullet$ by the functor $^\sim : D(A) \to D(\mathcal{O}_U)$.*
(2) *There is an affine open covering $X = \bigcup U_i$, $U_i = \mathrm{Spec}(A_i)$ such that for each i there exists a dualizing complex $\omega_i^\bullet$ for A_i such that $K|_{U_i}$ is isomorphic to the image of $\omega_i^\bullet$ by the functor $^\sim : D(A_i) \to D(\mathcal{O}_{U_i})$.*

Proof. Assume (2) and let $U = \mathrm{Spec}(A)$ be an affine open of X. Since condition (2) implies that K is in $D_{QCoh}(\mathcal{O}_X)$ we find an object $\omega_A^\bullet$ in $D(A)$ whose associated complex of quasi-coherent sheaves is isomorphic to $K|_U$, see Derived Categories of Schemes, Lemma 06Z0. We will show that $\omega_A^\bullet$ is a dualizing complex for A which will finish the proof.

Since $X = \bigcup U_i$ is an open covering, we can find a standard open covering $U = D(f_1) \cup \ldots \cup D(f_m)$ such that each $D(f_j)$ is a standard open in one of the affine opens U_i, see Schemes, Lemma 01IW. Say $D(f_j) = D(g_j)$ for $g_j \in A_{i_j}$. Then $A_{f_j} \cong (A_{i_j})_{g_j}$ and we have

$$(\omega_A^\bullet)_{f_j} \cong (\omega_i^\bullet)_{g_j}$$

in the derived category by Derived Categories of Schemes, Lemma 06Z0. By Dualizing Complexes, Lemma 0A7G we find that the complex $(\omega_A^\bullet)_{f_j}$ is a dualizing complex over A_{f_j} for $j = 1, \ldots, m$. This implies that $\omega_A^\bullet$ is dualizing by Dualizing Complexes, Lemma 0A7H. □